Republic of the Philippines
PARTIDO STATE UNIVERSITY
Research and Development Division
Camarines Sur

CERTIFICATION

To whom it may concern,

This is to certify that **Dr. Joni Neil B. Capucao** of Partido State University, College of Arts and Sciences has completed research approved by the University with the following details to wit:

Research Title	Proponent(s)	Original Approved Duration	Date of Completion	Authorship (Main Author, Co-Author)
Project Title: Implementation of progressive web application as platform for internal document warehousing: The effects of using web application in dynamic business process				
Study 1. Determining factors affecting submission, reviewing, and monitoring for internal document warehousing	Kennedy C. Cuya Dr. Joni Neil B. Capucao	May 2021-April 2022	June 9, 2022	Co- Author

This certification is issued for what ever reference/purpose it may served to the aboved requestor.

Issued this 9th day of June 2022 at Partido State University Research and Development Division.

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Director, Research and Development

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Vice President for Research, Extension and Knowledge Management

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DATE: 6/15/22

Determining factors affecting submission, reviewing, and monitoring for internal document warehousing

Partido State University
College of Arts and Sciences

Kennedy C. Cuya
Joni Neil B. Capucao

2022

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RESEARCH STUDY TERMINAL REPORT

BASIC INFORMATION

Study Title	: Determining factors affecting submission, reviewing, and monitoring for internal document warehousing
Researcher (s)	: Kennedy C. Cuya (Study Leader), Joni Neil B. Capucao
Campus/College	: College of Arts and Sciences Partido State University
Approved Budget	: Php600.00
Duration	: May 1, 2021 to April 30, 2022

EXECITIVE SUMMARY:

The changing technological pressure on management has led to an increasing interest in improving operational processes not merely to enhance performance but as well as staying current. Government organizations in the Philippines, particularly in the academe sector, such as the State Universities and Colleges, have experienced various management difficulties in implementing efficient automated enabled technologies. In response to these needs, the development of a web-based system for internal document warehousing system will enable the SUCs to automate the existing manual processes in document management, become more transparent, and provide efficient public service.

Partido State University (PSU), a state university from the Fourth Congressional District of Camarines Sur, Philippines has the same concern in following government mandates for a quality management system. However, most of the execution of processes in the University are manually done that often led to some specific problems like difficulty in tracking documents. The absence of a web-based internal document warehouse system in the different responsibility units of the University creates this difficulty.

The development of internal document warehousing system for the Partido State University is but a responsive, professional commitment of the proponent in addressing this gap.

The general objective of the present study is to design and develop the prototype internal document warehousing system to enhance the tracing of document's whereabouts for Partido State University, Goa, Camarines Sur. To address this need, the researcher answered the following questions. 1.) What system requirements (hardware, software, peopleware) can be adapted in establishing internal document warehouse; 2) What web services can be engineered that will serve as the basis for developing frameworks of the project(s); 3) What are the models that can be developed and applied to create the progressive web application.

The developed prototype internal document warehouse system was first subjected, through preliminary interviews with twenty-eight (28) office heads, to draw out the information requirements for the system's essentials that could adapt to the present demands. Specifically, they identified four (3) data requirements and three (3) information and communication technology (ICT) requirements. Along with the highest number of controlled documents regularly handled, Registrar and School of Graduate Studies

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have fifty-four (54) controlled documents. Along with the highest number of master lists of logbooks usually accomplished, the Registrar had twenty (20) logbooks. Nevertheless, the Registrar showed sixteen thousand four hundred forty-one (16,441), or 15.19 percent monitored documents. In terms of the information and communication technology requirements, there were three major components identified (system users, software, and hardware requirements).

The evaluations of information resources were now established. The researcher then designed diagrams or models for the validation of the eleven (11) member technical circle whose composition included information technology (IT) and software user experts. In terms of activity diagram, all eleven (11) or 100 percent jurors affirmed that there is one main activity needed. In terms of use case diagram, all eleven (11) or 100 percent jurors affirmed that five (3) system users (admin, stakeholders, staff/unit heads) were required. In terms of the deployment diagram, still, all eleven (11) or 100 percent jurors affirmed that use of Internet Browsers and Email were indispensable in the system features.

The developed internal document warehousing system was assessed by the end-users. As regard to usability level the following ratings were obtained: for usefulness, Mw 4.40 interpreted Very Good; for ease of use, Mw 4.37 interpreted Very Good; for ease of learning, Mw 4.44 interpreted Very Good; and for satisfaction, Mw 4.42 interpreted Very Good. Test of significant difference under usability level indicated the following p-values: 0.3806 for usefulness; 0.2627 for ease of use; 0.3818 for ease of learning; and 0.1123 for satisfaction, at .05 level of significance showed all four (4) indicators not significant.

Some information requirements were identified. These were the list of controlled documents, the master list of documents, monitored documents, code structures, peopleware, software, and hardware requirements. These data were identified as indispensable information requirements by the twenty-eight (28) office heads that it should be adopted in the design for internal document warehousing system so that the new system could address their generic needs.

There were three (3) models developed for PSU in internal document warehousing system – activity diagram, use case diagram, and deployment diagram. These features were chosen by technical experts and expert users within the local context of PSU who actually have direct involvement and hence, their contributory stance is overriding.

The respondents were satisfied with the developed internal document warehousing system. Tests of significant difference for usefulness, ease of use, ease of learning, and satisfaction level showed no significant difference and therefore respondents confirmed that the developed system met the basic requirements of the users. The internal document warehousing system, not only conformed to the generic needs of the staff's users, but also offered to the whole students, faculty, and other clientele credible and transparent transactions.

Some management concerns that are not part of document tracking and aggregating of expenses relative to information requirements should be attended later after the regular operation of the proposed prototype system.

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EDLYNDA M. PADILLA, PhD
DATE: 8/15/23

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Figure 18

Significance Difference in the Usability Level



5.0 Conclusion and Recommendation

Conclusions

Based on the findings, the conclusions below are presented.

1. Some information requirements were identified. These were the list of controlled documents, the master list of documents, monitored documents, peopleware, software, and hardware requirements. These data were identified as indispensable information requirements by the twenty-eight (28) office heads that it should be adopted in the design for internal document warehouse so that the new system could address their generic needs.

2. There were two (2) models developed for PSU in internal document warehouse – activity diagram and deployment diagram. These features were chosen by technical experts and expert users within the local context of PSU who actually have direct involvement and hence, their contributory stance is overriding.

3. The respondents were satisfied with the developed internal document warehouse. Tests of significant difference for usefulness, ease of use, ease of learning, and satisfaction level showed no significant difference and therefore respondents confirmed that the developed system met the basic requirements of the users. The prototype system, not only conformed to the generic needs of the staff's users, but also offered to the whole students, faculty, and other clientele credible and transparent transactions.

Recommendations

Based on the conclusions, the recommendations below are presented.

1. Some management concerns that are not part of internal document warehouse relative to information requirements should be attended later after the regular operation of the internal document warehouse system.

2. The research study came into existence as a result of institutional research requirement. In such a case, time constraints hindered some possible features such as the printing of tracking numbers using thermal printers that can be incorporated. This aspect may be given due attention when necessary.

3. There is a need for the developed internal document warehouse system to try-out an expanded service deliverance that could cover other stakeholders' transactions.

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INTERNAL DOCUMENT WAREHOUSING

PROJECT TITLE

IMPLEMENTATION PLAN

Objective:
To design and implement an internal document warehousing system to efficiently centralize, organize, and manage critical documents, ensuring they're easily accessible to authorized users and safeguarded from potential threats.

1. **Preliminary Analysis & Data Audit (June 10 - June 23)**
   - Catalogue existing document repositories and their types.
   - Assess the volume and format of current documents.
   - Determine the different user groups and their access levels.
   - Identify redundant or outdated documents for potential purging.
2. **Define Warehousing Requirements (June 24 - June 29)**
   - Determine storage capacity needs.
   - Understand desired features (e.g., search capabilities, versioning).
   - List security and backup needs.
3. **Hardware and Software Procurement (June 30 - July 9)**
   - Determine whether in-house hosting or a cloud solution is optimal.
   - Procure necessary servers, databases, and software.
   - Obtain the required licenses.
4. **System Design & Configuration (July 10 - July 30)**
   - Design the warehousing architecture, ensuring scalability.
   - Set up metadata tagging standards for efficient document retrieval.
   - Define user roles and access permissions.
5. **Document Migration & Organization (July 31 - August 19)**
   - Migrate documents from their original locations to the new warehouse.
   - Organize documents using the metadata tagging system.
   - Establish a folder hierarchy and naming conventions.
6. **Testing Phase (August 20 - August 27)**
   - Beta test with a select group of users.
   - Check for any access, retrieval, or storage issues.
   - Collect feedback and adjust the system accordingly.
7. **Training & Onboarding (August 28 - September 4)**
   - Train key personnel on how to upload, organize, and retrieve documents.
   - Distribute user manuals and FAQs.
   - Conduct workshops or webinars if necessary.
8. **Official Launch (September 5)**
   - Officially roll out the warehousing system to all users.
   - Ensure that a helpline or support system is in place.

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9. Continuous Monitoring & Updates (September 6 onwards)

- Monitor system usage and address any arising issues.
- Ensure regular backups.
- Continuously update software and hardware as needed.

GANTT CHART OF ACTIVITIES

Activity	June	July	August	September
Preliminary Analysis & Data Audit				
Define Warehousing Requirements				
Hardware & Software Procurement				
System Design & Configuration				
Document Migration & Organization				
Testing Phase				
Training & Onboarding				
Official Launch				
Continuous Monitoring & Updates				

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FACULTY POSITION RECLASSIFICATION FOR SUCs
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CERTIFICATION OF PERCENTAGE CONTRIBUTION

(Research Output Translated into Project/Policy or Product with Multiple Authors)

Title of Research: **Determining factors affecting submission, reviewing and monitoring for internal document warehousing**

Title of Project/Policy/Product **Internal Document Warehousing**

Instruction: Supply ALL the names of the authors involved in the research output translated inyp project/policy or product and indicate the contribution of each author in percentage. Each author shall sign the conforme column if he/she agrees with the distibution. The conforme should be signed by all the authors in order to be considered. Please prepare a separate Certification for each output.

	Name	Current Affiliation	% Contribution	Conforme *(Sign if you agree with the % distribution)*
1	Kennedy C. Cuya	Partido State University	60%	
2	Dr. Joni Neil B. Capucao	Partido State University	40%	
3				
4				
5				
6				
7				

**should have a total of 100%* 100%

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